FINITE TRUNCATIONS FROM REPLACEMENT, AND A 3-TRUNCATION OF S^2

Purpose and status. The informal proof for the construction of finite truncations in Unit and for an inhabitant of the interface `Trunc3` of `sag/pi3-s2-postnikov` (see `doc/pi3-s2-postnikov.pdf`). It was written before the formal proof, and each lemma below is a declaration (§7). It was refereed independently before formalisation: no mathematical error was found, and the corrections asked for (universe levels, cited dependencies, order of generalisation before J) are included below. The formal proof is checked: `SphereTruncation.SphereTrunc.trunc3` is an inhabitant of `Trunc3`, from the sphere axiom and the `Theory` assumptions, with no further assumption. Foundations: the `Theory` assumptions (univalence, images, surjectivity of the image maps; function extensionality derived); for §5 also the sphere axiom. No other axiom. The construction is Rijke's (*The join construction*, §7): $\|A\|_{n+1}$ is the image of $a \mapsto \|a = -\|_n$. Here the join is not used. Replacement provides the small image, and the inductive definition of connectedness replaces the connectivity theorems of the paper.

1. Conventions

Unit indexing: $\mathsf{isTrunc}\, m\, X$ is "X is $(m-2)$-truncated" ($\mathsf{isTrunc}\, 0$ is contractible, $\mathsf{isTrunc}\,(m+1)\, X$ means all identity types are $\mathsf{isTrunc}\, m$); $\mathsf{isConn}\, m\, X$ is "X is $(m-2)$-connected" ($\mathsf{isConn}\, 0 = \top$, and $\mathsf{isConn}\,(m+1)\, X$ means $\|X\|_{-1}$ and all identity types are $\mathsf{isConn}\, m$). Both are propositions (`trunc-prop`, `conn-prop`). A proposition is $\mathsf{isTrunc}\, m$ for every $m \geq 1$ (`prop-trunc`), and $\mathsf{isConn}\,(m+1)$ implies $\mathsf{isConn}\, m$ (`conn-down`). $\mathrm{fib}_f(b) = \sum_a (f\, a = b)$. Everything lives in one universe $U = U_l$.

2. Connected maps and truncated families

Lemma 1 (null types)**.** *For every m, if $\mathsf{isConn}\, m\, C$ and $\mathsf{isTrunc}\, m\, D$, then $\mathrm{const} : D \to (C \to D)$ is an equivalence.*

Proof. Induction on m. For $m = 0$, D and $C \to D$ are contractible. For $m+1$: being an equivalence is a proposition and C is merely inhabited, so let $c_0 : C$. Put $\mathrm{ev}\, g = g\, c_0$; then $\mathrm{ev}(\mathrm{const}\, d) = d$ by definition. For $g : C \to D$ and $c : C$, the type $c = c_0$ is $\mathsf{isConn}\, m$ and $g\, c = g\, c_0$ is $\mathsf{isTrunc}\, m$. By the induction hypothesis there is $h_c : g\, c = g\, c_0$ with $\mathrm{const}\, h_c = \mathsf{ap}_g$, namely the inverse of const at $\mathsf{ap}_g : (c = c_0) \to (g\, c = g\, c_0)$. So $\mathrm{const}(\mathrm{ev}\, g) = g$ by function extensionality with h_c^{-1}. Hence const has the two-sided inverse ev. □

Lemma 2 (elimination along a connected map)**.** *Let $f : A \to B$ with $\mathsf{isConn}\, m\, (\mathrm{fib}_f\, b)$ for all b, and $P : B \to U$ with $\mathsf{isTrunc}\, m\, (P\, b)$ for all b. Then precomposition $f^* : \prod_b P\, b \to \prod_a P(f\, a)$ is an equivalence.*

Proof. Let $\Psi : \prod_b (\mathrm{fib}_f\, b \to P\, b) \to \prod_a P(f\, a)$, $\Psi\, t\, a = t\, (f\, a)\, (a, \mathsf{refl})$. It has the inverse $\Phi\, s\, b\, (a, p) = p_*(s\, a)$: $\Psi\Phi = \mathrm{id}$ by definition, $\Phi\Psi = \mathrm{id}$ by J on p (and function extensionality). Let $K : \prod_b P\, b \to \prod_b (\mathrm{fib}_f\, b \to P\, b)$ be $K\, s\, b = \mathrm{const}(s\, b)$. It is an equivalence: it is a Π of equivalences (Lemma 1), and a Π of equivalences is an equivalence by function extensionality. Finally $f^* = \Psi \circ K$ by definition. □

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In particular (constant P): for $\mathsf{isTrunc}\, m\, Y$, $f^* : (B \to Y) \to (A \to Y)$ is an equivalence; and maps $B \to Y$, or sections, that agree after f^* are equal.

Lemma 3 (fibres of ap)**.** *For $f : A \to B$, $x, y : A$ and $w : f\,x = f\,y$,*

$$\mathrm{fib}_{\mathsf{ap}_f}(w) \;\simeq\; \big((x, w) = (y, \mathsf{refl})\big) \quad \textit{in } \mathrm{fib}_f(f\,y).$$

Hence if all fibres of f are $\mathsf{isConn}\,(m+1)$, all fibres of $\mathsf{ap}_f : (x = y) \to (f\,x = f\,y)$ are $\mathsf{isConn}\, m$.

Proof. Paths in a Σ-type: $((x, w) = (y, \mathsf{refl})) \simeq \sum_{q : x = y} (q_* w = \mathsf{refl})$. Moreover $q_* w = (\mathsf{ap}_f q)^{-1} \cdot w$ (`Groupoid.transport-fib`), and $(\alpha^{-1} \cdot w = \mathsf{refl}) \simeq (\alpha = w)$. □

Lemma 4 (connectivity along a connected map)**.** *If all fibres of $f : A \to B$ are $\mathsf{isConn}\, m$ and $\mathsf{isConn}\, m\, A$, then $\mathsf{isConn}\, m\, B$.*

Proof. Induction on m; $m = 0$ is trivial. For $m + 1$: B is merely inhabited because A is. For b, b' the goal $\mathsf{isConn}\, m (b = b')$ is a proposition and the fibres over b, b' are merely inhabited, so we may assume $b = f\,a$, $b' = f\,a'$. Then apply the induction hypothesis to $\mathsf{ap}_f : (a = a') \to (f\,a = f\,a')$: its fibres are $\mathsf{isConn}\, m$ by Lemma 3, and $a = a'$ is $\mathsf{isConn}\, m$. □

3. The truncations

Definition 5 (truncation structure at level m)**.** T_m assigns to each $A : U$ a type $\mathsf{T}_m A : U$ and a map $|-| : A \to \mathsf{T}_m A$ with $\mathsf{isTrunc}\, m\, (\mathsf{T}_m A)$ and $\mathsf{isConn}\, m\, (\mathrm{fib}_{|-|}\, z)$ for all z.

By Lemma 2 a truncation structure has dependent elimination into $\mathsf{isTrunc}\, m$ families, uniquely. For $m \geq 1$ this includes families of propositions.

Lemma 6 (base, $m = 1$)**.** *The propositional truncation (`Truncation`) is a truncation structure at level 1: it is a proposition, and each fibre of its unit is merely inhabited, by induction into propositions.*

$\mathsf{T}_0 A = \top$ is also a truncation structure, but the step below is not proved from $m = 0$: it eliminates into families of propositions, which are $\mathsf{isTrunc}\, m$ only for $m \geq 1$. So the recursion starts at $m = 1$.

The step. Let T_m be a truncation structure, $m \geq 1$, and $A : U$. Let $f : A \to (A \to U)$, $f\,a\,c = \mathsf{T}_m (a = c)$. The type $A \to U$ is locally small, since $(g = h) \simeq \prod_c (g\,c \simeq h\,c)$ by function extensionality and univalence (`Quotient.FunctionSmall`). So replacement gives the image I of f: a small type, a surjection $q : A \to I$, an embedding $i : I \to (A \to U)$ and $t_a : i(q\,a) = f\,a$. Define $\mathsf{T}_{m+1} A = I$ and $|a| = q\,a$.

Lemma 7 (Yoneda for truncated families)**.** *For $a : A$ and a family $Q : A \to U$ with $\mathsf{isTrunc}\, m (Q\,c)$ for all c, the map*

$$\mathrm{Y} : \prod_c (\mathsf{T}_m (a = c) \to Q\,c) \to Q\,a, \qquad \Phi \mapsto \Phi\,a\,|\mathsf{refl}|$$

is an equivalence. Its inverse sends v to the family $\Phi_v\,c =$ the extension along $|-|$ of $p \mapsto p_* v$.

Proof. Y is the composite of $\prod_c |-|^*$, which is an equivalence for each c by Lemma 2 with $Q\,c$ constant, and of the evaluation at refl, $\prod_c ((a = c) \to Q\,c) \to Q\,a$. The latter is an equivalence with inverse $v \mapsto (c, p) \mapsto p_* v$, by J on p and function extensionality. □

Lemma 8 (all such families are equivalences)**.** *For $a, b : A$, every $\Phi : \prod_c (\mathsf{T}_m (a = c) \to \mathsf{T}_m (b = c))$ is a family of equivalences.*

Proof. By Lemma 7 with $Q\,c = \mathsf{T}_m(b = c)$, $\Phi = \Phi_v$ with $v = \mathrm{Y}\Phi : \mathsf{T}_m(b = a)$ (the counit of the equivalence, a path, along which the property is transported). So it suffices to prove, for all a and all $v : \mathsf{T}_m(b = a)$, that $\Phi_v^{(a)}\,c$ is an equivalence for all c. This is a proposition in v, so by elimination (here $m \geq 1$ is used) we may take $v = |r|$ with $r : b = a$. By J on r, with a free, take $a \equiv b$ and $r = \mathsf{refl}$. Then $\Phi_{|\mathsf{refl}|}\,c$ extends $p \mapsto p_*|\mathsf{refl}| = |p|$ (by J on p), as the identity of $\mathsf{T}_m(b = c)$ does. By uniqueness of extensions it is the identity, an equivalence. □

Proposition 9 (identity types)**.** $(q\,a = q\,b) \;\simeq\; \mathsf{T}_m(b = a)$ *for all* $a, b : A$.

Proof. $(q\,a = q\,b) \simeq (i(q\,a) = i(q\,b))$ because i is an embedding; $\simeq (f\,a = f\,b)$ by conjugation with t_a, t_b; $\simeq \prod_c (f\,a\,c \simeq f\,b\,c)$ (`FunctionSmall`); $\simeq \prod_c (f\,a\,c \to f\,b\,c)$. The last step holds because $\prod_c (X_c \simeq Y_c)$ is $\sum_\Phi \prod_c \mathsf{isEquiv}(\Phi\,c)$, a Σ of propositions that all hold (Lemma 8). Finally $\simeq \mathsf{T}_m(b = a)$ by Lemma 7. □

Proposition 10 (level)**.** $\mathsf{isTrunc}\,(m+1)\,I$.

Proof. For $z, w : I$ the goal $\mathsf{isTrunc}\,m(z = w)$ is a proposition. Since q is surjective (induction into propositions) we may assume $z = q\,a$, $w = q\,b$. Then use Proposition 9 and $\mathsf{isTrunc}\,m(\mathsf{T}_m(b = a))$. □

Proposition 11 (connectivity of the unit)**.** $\mathsf{isConn}\,(m+1)\,(\mathrm{fib}_q\,z)$ *for all* $z : I$.

Proof. A proposition in z; so we may assume $z = q\,b$. By Proposition 9 (a Σ of equivalences), $\mathrm{fib}_q(q\,b) \simeq S_b := \sum_a \mathsf{T}_m(b = a)$, and isConn is invariant under equivalence. S_b is merely inhabited by $(b, |\mathsf{refl}|)$. For $(a, u), (a', u') : S_b$ the goal $\mathsf{isConn}\,m((a, u) = (a', u'))$ is a proposition in u. By elimination take $u = |p|$, $p : b = a$, and by J on p, with a free, take $(a, u) = (b, |\mathsf{refl}|)$. Then

$$((b, |\mathsf{refl}|) = (a', u')) \;\simeq\; \textstyle\sum_{r:b=a'} (r_*|\mathsf{refl}| = u') \;\simeq\; \sum_{r:b=a'} (|r| = u') \;=\; \mathrm{fib}_{|-|}(u'),$$

by paths in a Σ-type and $r_*|\mathsf{refl}| = |r|$ (J on r). This is $\mathsf{isConn}\,m$ because T_m is a truncation structure. □

Theorem 12. *For every $m \geq 1$ there is a truncation structure T_m, by recursion on m from Lemma 6 and the step (Propositions 10, 11).*

4. Pointed universal property and π_3

Let $\eta = (|-|, \mathsf{refl}) : (X, x_0) \to_* (\mathsf{T}_m X, |x_0|)$.

Proposition 13 (pointed universal property)**.** *For* $\mathsf{isTrunc}\,m\,Y$, $\eta^* : \mathrm{Map}_*(\mathsf{T}_m X, Y) \to \mathrm{Map}_*(X, Y)$ *is an equivalence.*

Proof. $\mathrm{Map}_*(\mathsf{T}_m X, Y) = \sum_g (g|x_0| = y_0)$, and $(|-|^* g)\,x_0$ is $g|x_0|$ by definition. So the Σ over the equivalence $|-|^*$ of bases (Lemma 2) is an equivalence onto $\sum_h (h\,x_0 = y_0)$. It agrees with η^* up to $\mathsf{refl} \cdot w = w$ in the pointing. □

Lemma 14 (loopings of connected maps)**.** *If* $F : (Y, y_0) \to_* (Z, z_0)$ *has all fibres* $\mathsf{isConn}\,(n+1)$, *then* $\Omega F : \Omega Y \to \Omega Z$ *has all fibres* $\mathsf{isConn}\,n$.

Proof. ΩF is homotopic to $q \mapsto F_0^{-1} \cdot \mathsf{ap}_F q \cdot F_0$ (`loop-ap`); by J on the pointing F_0 we may take $F_0 = \mathsf{refl}$, and then ΩF is ap_F up to $\mathsf{refl} \cdot -$. Here ΩF is the pointed map `Maps.loop F`, and $\Omega^3\eta$ below is `Maps.loop (loop2 eta)`, the map that `PiOne.Induced` uses for $\pi_3\eta$. Fibres of homotopic maps are equivalent (`fib-htpy`). Conclude by Lemma 3. □

Proposition 15 (π_3)**.** *For* $m = 5$ *(3-types),* $\pi_3\eta : \pi_3 X \to \pi_3(\mathsf{T}_5 X)$ *is bijective.*

Proof. The fibres of $|-|$ are $\mathsf{isConn}\,5$. By Lemma 14 three times, $\Omega^3\eta$ has fibres $\mathsf{isConn}\,2$: merely inhabited, with any two points merely equal. $\pi_3\eta$ is the map induced by $\Omega^3\eta$ on the quotients of $\Omega^3 X$ and $\Omega^3\mathsf{T}_5 X$ by mere equality. It is surjective: by induction on the class and the mere inhabitation of fibres. It is injective: if $\Omega^3\eta\, q$ and $\Omega^3\eta\, q'$ are merely equal (effectiveness of the quotient by mere equality, `Quotient.effective`; both classes by induction into a proposition), then (q, p) and (q', refl) lie in one fibre, so they are merely equal, so q and q' are. A map of sets with propositional, merely inhabited fibres has contractible fibres, so it is an equivalence. No choice is used. □

5. The 3-truncation of S^2

Lemma 16. $\mathsf{isConn}\,3\,S^2$ *(simply connected).*

Proof. Let $V = \mathsf{T}_3 S^2$, a 1-type. (a) For any 1-type Y, evaluation at base, $(S^2 \to Y) \to Y$, is an equivalence: it is the first projection after the axiom ev_2, and the fibres $\Omega^2(Y, y)$ of the projection are contractible: $y = y$ is $\mathsf{isTrunc}\,2$, so $\Omega^2(Y, y)$ is a proposition, inhabited by refl. (b) The maps id and const $|\mathsf{base}|$ of V agree after $|-|^*$. Indeed $|-|$ and const $|\mathsf{base}|$ agree at base by definition, so they are equal by (a) with $Y = V$. By the universal property of V (Lemma 2), id = const $|\mathsf{base}|$, so V is contractible. (c) Then $\mathrm{fib}_{|-|}(|\mathsf{base}|) \simeq S^2$: a Σ of contractible types ($|s| = |\mathsf{base}|$). The fibre is $\mathsf{isConn}\,3$, hence so is S^2. □

Theorem 17 (an inhabitant of `Trunc3`)**.** *Take $T = \mathsf{T}_5 S^2$, $t_0 = |\mathsf{base}|$ and η as above. Then* $\mathsf{isTrunc}\,5\,T$ *(Theorem 12);* $\mathsf{isConn}\,3\,T$ *by Lemma 4, since the fibres of $|-|$ are* $\mathsf{isConn}\,5$*, hence* $\mathsf{isConn}\,3$ *(`conn-down`), and S^2 is* $\mathsf{isConn}\,3$ *(Lemma 16);* ***`up`*** *is Proposition 13;* ***`pi3-eta`*** *is Proposition 15.*

6. Universes

All the constructed types are in U_l. The image is taken with codomain $A \to U_l : U_{l+1}$, which is locally small: its identity types are *equivalent* to types of U_l (`Small` is a type with an equivalence). This uses function extensionality at U_{l+1} (in `Quotient.FunctionSmall`), derived from univalence at $l + 1$, and univalence at l. Replacement is used as `Images` and `Surjectivity` at $(l, l + 1)$, eliminating into U_l. These are instances of the `Theory` parameters, which are level-polymorphic. S^2 and its axiom are at level l, and the targets Y of its universal property are in U_l. In particular V and T are, as is the target of every elimination.

7. Correspondence with the formal declarations

Lemma 1	`ConnectedMaps.null`
Lemma 2	`ConnectedMaps.Elim`: `Psi`, `Phi`, `K`, `precomp-def`, `equiv`, `ext`, `ext-beta`, `unique`
Lemma 3	`ConnectedMaps.fib-ap`, `fib-ap-conn`
Lemma 4	`ConnectedMaps.conn-up`
truncation structure	`Truncations.TruncOp`
Lemma 6	`Truncations.base`
Lemma 7	`Truncations.Step.Yon`: `Y`, `equiv`, `Phi-of`, `Phi-of-Y`
Lemma 8	`Step.At.AllEq.all-equiv` (`id-at`, `good-unit`, `good`)
Proposition 9	`Step.At.Ident`: `e1`–`e4`, `ident`
Proposition 10	`Step.At.trunc'`
Proposition 11	`Step.At.ConnS`, `Step.At.conn'`
Theorem 12	`Truncations.TO` (locked)
Proposition 13	`TruncationProps.PtUP.up`
Lemma 14	`TruncationProps.loop-conn`
Proposition 15	`TruncationProps.ConnPi1.equiv`
Lemma 16	`SphereTruncation.SphereTrunc`: `ev-base-equiv`, `V-contr`, `s2-conn`
the inhabitant of `Trunc3`	`SphereTrunc.trunc3`

Checking against the text (`seg/WRITING.md`). `truncation.opens` declares, for each of the 94 declarations of these four files, the definitions its step uses. These are the notions of its statement and the components named in the text. The uniform checker (`-opens`) reports nothing else opened, except in three places: (i) `ConnPi1` opens the carriers of the π_1 quotient (`PiOne.grp`, `QGrp.G`, `Quot.Q`, `image`), an interface issue of the shared library, which was not edited; (ii) `c2` opens `Maps.D`: the base point of a looping is `refl` by computation; (iii) `V-contr`, `c5`, `connT` report `TO` and the image of the step as opened, although `TO` is locked. A test that needs `TO` to unfold is rejected by both checkers, so no conversion relies on it. The cause of this report was not determined. `TO`, `null`, `Elim.equiv` and `Elim.ext` are locked after their characterising lemmas. Every declaration checks in under 0.01 s.